

Global trace integrality and the original Erdős–Straus source

The Clankers*

20 September 2026

Abstract

For $n \geq 3$ rational nonzero denominators with reciprocal sum $4/p$, the first $n - 1$ integral power sums force all denominators to be integers. We prove the more general common-denominator bound $d^n \mid (n - 1)!m$ for reciprocal numerator m , and classify the exact possible denominators at $n = 3$. On a complete unfiltered integer ES gate source, a single literal trace recovers the full original divisor budget and the channel. Its two-point fibres are exactly the original middle orientations. An integral first trace alone on arbitrary rational tails leaves a precisely computed square-part obstruction. Finally, for every hard prime we construct positive irrational algebraic-integer targets with all integer traces, the original middle prime-local signature, and the inherited scalar height bounds. Thus the rationality domain and the global integrality observation are both indispensable. No universal ES occupancy statement is asserted.

1 Scope and the exact arithmetic return

The preceding integral-completion calculation retained the original finite orders, but produced rational numerical candidates which pass a specified finite set of local tests while violating one original prime-exponent bound [4, §“One unavailable prime occurrence”]. We now test a global observation instead. For rational ES denominators, two integral Newton traces force all denominators to be integers. In the original raw integer gate source, one integral trace already forces the correct divisor box, with a necessary middle-to-exterior reclassification at the distinguished prime. The complete trace fibres recover the middle states as actual kernel generators.

This is an arithmetic integrality theorem, not an assertion that an integral trace is attained at every prime. The remaining universal existence question is stated at the end. In particular, an algebraic-integer root is not silently identified with a rational integer. We construct an explicit algebraic counterfamily at every hard prime which retains all integral power traces and the middle prime-local signature, but has no rational denominator pair. No historical-priority determination is claimed.

2 A denominator theorem for finite rational spectra

Let $x_1, \dots, x_n \in \mathbb{Q}^\times$, $n \geq 2$. Write

$$s_k = \sum_{i=1}^n x_i^k, \quad \sum_{i=1}^n \frac{1}{x_i} = \frac{m}{b} \neq 0, \quad b > 0, \quad \gcd(m, b) = 1.$$

The numerator m is signed; divisibility of it means divisibility of $|m|$. The roots are labelled and may repeat.

*Research continuation, with exact arithmetic certificates. The stated verification is not independent mathematical review.

Theorem 2.1 (Common denominator and reciprocal numerator). *If $s_1, \dots, s_{n-1} \in \mathbb{Z}$, the reduced denominators of all x_i are the same integer d . Moreover*

$$\boxed{\gcd(d, b) = 1, \quad d^n \mid (n-1)!m.} \quad (\text{TR1})$$

If n is odd, d is odd.

Proof. Let e_j be the elementary symmetric functions, with $e_0 = 1$. The exact Newton identities are

$$je_j = \sum_{k=1}^j (-1)^{k-1} e_{j-k} s_k. \quad (\text{TR2})$$

They follow by expanding the logarithmic derivative of $\prod_i (1 + x_i T)$ as a formal power series and comparing the coefficient of T^{j-1} . This is the classical identity; compare Euler [1, §§3,5–7]. Induction in TR2 proves $j!e_j \in \mathbb{Z}$ for $j < n$: multiplying its right side by $(j-1)!$ clears all previous denominators.

Fix a prime ℓ at which some x_i is nonintegral. Let $-a < 0$ be their minimum valuation, and let exactly r roots have this valuation. If $r < n$, the product of those r roots is the unique least-valuation term in e_r , so $v_\ell(e_r) = -ar$. But

$$v_\ell(e_r) \geq -v_\ell(r!) > -r \geq -ar,$$

a contradiction. The strict factorial bound follows from $\sum_{k \geq 1} \lfloor r/\ell^k \rfloor < r$ (or its finite base- ℓ digit formula). Therefore every root has the same valuation $-a$ at that prime. This proves equality of all reduced denominators.

Write $x_i = A_i/d$, where each $\gcd(A_i, d) = 1$. Since $e_{n-1}/e_n = m/b$ and $e_n = \prod_i A_i/d^n$, integrality of $(n-1)!e_{n-1}$ gives at every $\ell^a \parallel d$

$$na \leq v_\ell((n-1)!) + v_\ell(m) - v_\ell(b). \quad (\text{TR3})$$

The factorial valuation is less than n . Hence $\ell \mid m$; reducedness then gives $\ell \nmid b$. Equation TR3 proves TR1. Finally, if d were even and n odd, all reduced numerators would be odd, and their sum would be odd, contradicting $s_1 \in \mathbb{Z}$. $\square$

Corollary 2.2 (Numerator four, any length at least three). *Let $n \geq 3$, $p \in \mathbb{Z}_{>0}$, and $x_i \in \mathbb{Q}_{>0}$ satisfy $\sum_i 1/x_i = 4/p$. Then*

$$\boxed{x_i \in \mathbb{Z} \ (1 \leq i \leq n) \iff s_1, \dots, s_{n-1} \in \mathbb{Z}.} \quad (\text{TR4})$$

For the Erdős–Straus equation, only s_1, s_2 are required. If p is odd, these also imply $s_2 \equiv s_1^2 \pmod{8}$.

Proof. The reduced reciprocal numerator divides four. An odd prime cannot divide d , since $v_\ell((n-1)!) < n$ in TR1. If n is odd, the final assertion of Theorem 2.1 also excludes two. If $n \geq 4$ is even, $n-1$ is odd and at least three. Counting the binary digits of $n-1$ gives

$$v_2((n-1)!) + 2 = n + 1 - s_2^{\text{bin}}(n-1) \leq n-1,$$

so TR1 again excludes two. Thus $d = 1$. The converse is literal. For three integer denominators, $4e_3 = pe_2$ and odd p give $4 \mid e_2$; use $s_2 = s_1^2 - 2e_2$. $\square$

The restriction $n \geq 3$ is necessary: $(x_1, x_2) = (p/2, p/2)$ at odd p has integer first trace and reciprocal sum $4/p$, but is not integral.

2.1 The exact possible denominators at length three

Theorem 2.3 (Sharp three-root denominator classification). *Under Theorem 2.1 with $n = 3$,*

$$\boxed{d \text{ is odd, } v_3(d) \leq 1, \quad \ell \mid d, \ell \neq 3 \implies \ell \equiv 1 \pmod{3}, \quad d^3 \mid m.} \quad (\text{TR5})$$

Conversely every d with these prime conditions is realized by three positive rationals with common reduced denominator d , integral s_1, s_2 , and $v_\ell(m) = 3v_\ell(d)$ at every $\ell \mid d$.

Proof. The oddness follows above, and TR1 then gives $d^3 \mid m$. At a denominator prime, scale the three roots to local units A, B, C . Their first and second symmetric functions vanish modulo ℓ , so their residues are the roots of $T^3 - ABC$. If $\ell \neq 3$, the polynomial has distinct nonzero roots. Ratios of these roots supply a nontrivial cube root of one, giving $\ell \equiv 1 \pmod{3}$. For $\ell = 3$ their residues coincide. If $a = v_3(d) \geq 2$, the exact symmetric identities imply

$$A^2 + AB + B^2 = 3^a s_1(A + B) - 3^{2a} e_2.$$

The right side has valuation at least two, while $4(A^2 + AB + B^2) = (2A + B)^2 + 3B^2$ has valuation exactly one. This is impossible. Other denominator primes are units in this displayed local equation.

The denominator $d = 1$ is realized by $(1, 2, 3)$. For $d > 1$, work independently at each $\ell^a \parallel d$ modulo ℓ^{2a+1} , taking the first numerator to be one. At $\ell = 3$, $a = 1$, take $(A, B, C) = (1, 1, 4) \pmod{27}$; the sum is divisible by three and the pair sum has valuation exactly two. At any other denominator prime choose a nontrivial cube root b modulo ℓ^{2a} . It is obtained from a root of $b^2 + b + 1$ modulo ℓ by the explicit unit-derivative lift

$$b' = b + \ell^k t, \quad t \equiv -\frac{b^2 + b + 1}{\ell^k} (2b + 1)^{-1} \pmod{\ell}.$$

Set $c = -1 - b + \ell^{2a} t$. Its pair sum is $-(b^2 + b + 1) + (1 + b)\ell^{2a} t$. Because $1 + b$ is a unit, precisely one value of $t \pmod{\ell}$ makes its valuation exceed $2a$; choose any other value. CRT now gives integer B, C with these separate labelled residues. Add respectively one and two multiples of $L = \prod_{\ell^a \parallel d} \ell^{2a+1}$ to make $1 < B < C$. All three numerators are coprime to d ; their sum is divisible by d and their pair sum has exact valuation $2a$ at each denominator prime. Dividing by d gives integral first two traces. The reciprocal numerator has exact valuation $3a$ because the product of the numerators is a unit there. $\square$

Examples retaining the exact reciprocal numerators are

(x_1, x_2, x_3)	(s_1, s_2)	$\sum_i 1/x_i$
$(1/3, 1/3, 4/3)$	$(2, 2)$	$27/4$
$(1/7, 18/7, 30/7)$	$(7, 25)$	$343/45$
$(1/13, 22/13, 146/13)$	$(13, 129)$	$10985/803$

(TR6)

The cube restriction concerns the reduced numerator, not the number of primes one has elected to check.

3 One global trace on an unfiltered integer gate source

Now fix prime $p \equiv 1 \pmod{4}$ and a complete original first-half shell

$$p/4 < a < p/2, \quad R = 4a - p.$$

Then $R \geq 3$, $R \equiv 3 \pmod{4}$, and $\gcd(pa, R) = 1$. Before imposing any divisor budget, take an arbitrary positive integer U with $R \mid a + U$. Define

$$\boxed{x = a, \quad y = \frac{pa(U + a)}{RU}, \quad z = \frac{p(a + U)}{R}, \quad S = p + x + y + z.} \quad (\text{TR7})$$

These are positive rational denominators, z is an integer, and direct multiplication gives $1/x + 1/y + 1/z = 4/p$.

Theorem 3.1 (Global trace recovers the original budget and tag). *The exact reduced denominator is*

$$\boxed{\text{den } S = \text{den } y = \frac{U}{\gcd(U, pa^2)}}. \quad (\text{TR8})$$

Consequently $S \in \mathbb{Z}$ if and only if $U \mid pa^2$. Its integral returns are exactly the disjoint tagged cases

$v_p(U)$	original divisor	channel
0	$u = U \mid a^2$	M
1	$u = U/p \mid a^2$	E .

(TR9)

No higher p -valuation gives an integral return. On an integral trace the tag can also be read without factoring U :

$$\boxed{M \iff S \equiv a \pmod{p}, \quad E \iff S \not\equiv a \pmod{p}}. \quad (\text{TR10})$$

For the raw E gate $R \mid 4U + 1$, the analogous rational return

$$\left(a, \frac{paU + a^2}{RU}, \frac{pa + p^2U}{R} \right)$$

has $\text{den } S = U / \gcd(U, a^2)$; its trace is integral exactly on the original E box.

Proof. The gate makes $a + U$ divisible by R , and $\gcd(U, R) = 1$. If $n = pa(U + a)/R \in \mathbb{Z}$, then, modulo U , $Rn \equiv pa^2$. Hence $\gcd(n, U) = \gcd(pa^2, U)$, proving TR8. For integral returns $a < p$ makes the p -part of U zero or one. At valuation zero, the original M gate is unchanged. At valuation one write $U = pu$. Since $p \equiv 4a \pmod{R}$, the gate $a + pu \equiv 0$ is exactly $4u + 1 \equiv 0$; the two expressions in TR7 are the original E denominators at u . They are not a canonical M state with an impermissible extra occurrence of p .

For completeness, the original normalization is

$$g = \gcd(a, u), \quad h = g^2/u, \quad r = u/g, \quad s = a/g.$$

Prime valuations give positive integral h, r, s , $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. At M set $\lambda = (r + s)/R$ and return $(a, phs\lambda, phr\lambda)$. At E set $\kappa = (pr + s)/R$ and return $(a, hsk, phr\kappa)$. The gates prove their integrality. The ordered inverses are $pa^2/(Ry - pa)$ and $a^2/(Ry - pa)$ respectively. These Type-I/II identities are inherited [2, §2, Propositions 2.2 and 2.6]. At M both tail denominators are multiples of p , so TR10 holds. At E the first tail denominator is a p -unit, proving the other alternative. The raw E calculation is identical: its third denominator is integral by multiplying the E gate by pa , and reduction of the second numerator modulo U gives a^2 . $\square$

For a nonintegral raw word, TR8 gives the exact integer-spacing estimate

$$\boxed{\text{dist}(S, \mathbb{Z}) \geq \frac{1}{\text{den } S} \geq \frac{1}{U}}. \quad (\text{TR11})$$

Strict proximity $|S - N| < 1/U$ to an integer forces equality. No estimate proving that proximity for some raw word has been obtained. The strict sign is necessary: at $p = 13, a = 4, R = 3, U = 5$, $S = 436/5$ and the distance is $1/5$.

At $p = 35281, a = 8821, U = 202883$, the preceding foreign-prime candidates have exact literal traces

$$S_E = 1936132532779018/23, \quad S_M = 59754044938/23. \quad (\text{TR12})$$

Thus the global trace rejects their one unavailable prime occurrence. This is a stronger *test* than the earlier fixed finite-place predicate, not a proof that the test succeeds at every prime.

4 The trace fibres and their integral coefficient kernel

Let

$$\Omega_{p,a} = \{1 \leq U \leq pa^2 : R \mid a + U\}, \quad \Theta_{p,a} = \{S(U) : U \in \Omega_{p,a}\},$$

and define the actual free-abelian pushforward

$$\Phi : \mathbb{Z}^{\Omega_{p,a}} \longrightarrow \mathbb{Z}^{\Theta_{p,a}}, \quad e_U \longmapsto e_{S(U)}. \quad (\text{TR13})$$

The raw source is finite without assuming any original hit. Its explicit size is $1 + \lfloor (pa^2 - U_0)/R \rfloor$, where $U_0 = (-a) \bmod R \in [1, R - 1]$.

Theorem 4.1 (Complete trace fibres). *Two distinct raw words have the same trace exactly when $UV = a^2$. Every such pair consists of the two original middle orientations. Every exterior integral trace has one raw preimage; every middle integral trace has two; every nonintegral trace has one. If E_a, M_a are the original labelled counts, with both middle orientations retained, then*

$$\boxed{\text{rank ker } \Phi = M_a/2, \quad |\Theta_{p,a} \cap \mathbb{Z}| = E_a + M_a/2, \quad E_a + M_a = |\Theta_{p,a} \cap \mathbb{Z}| + \text{rank ker } \Phi.} \quad (\text{TR14})$$

The kernel has the explicit basis $e_U - e_{a^2/U}$ for the middle words $U < a$.

Proof. From TR7,

$$S(U) - S(V) = \frac{p}{R}(U - V) \left(1 - \frac{a^2}{UV}\right). \quad (\text{TR15})$$

If both raw words are integers and $UV = a^2$, they are original square divisors. The gate at U implies the gate at V since $U \equiv -a \pmod{R}$ and is a unit. Thus both are M. There is no fixed point: $U = a$ would make $R \mid 2a$, impossible. Conversely every original M complement is present in $\Omega_{p,a}$. An E word has $v_p(U) = 1$; its complementary rational number a^2/U is not an integer, so its fibre is a singleton. These are the entire fibres, proving TR14 and the stated integral basis. $\square$

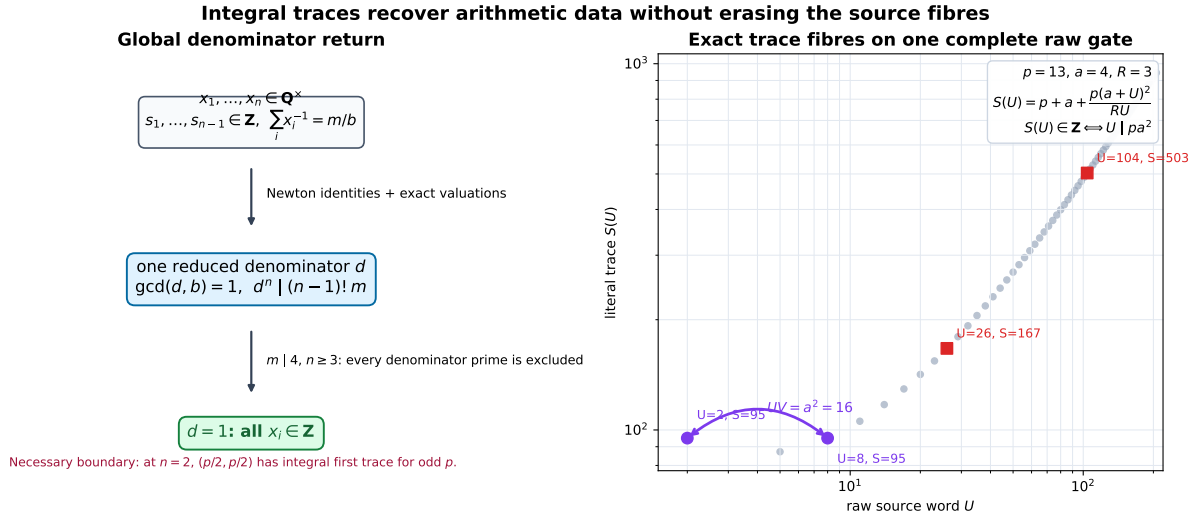


Figure 1: The common-denominator implication and one complete raw trace source. In the displayed $p = 13$, $a = 4$, $R = 3$ example, all raw words satisfying $R \mid a + U$ are retained. The integral points are precisely $U = 2, 8, 26, 104$, as TR8 predicts. The two middle words $U = 2, 8$ share $S = 95$ because their product is $a^2 = 16$; the exterior words are singleton fibres. The figure therefore exhibits the nonzero kernel generator $e_2 - e_8$ rather than identifying the two source words. All plotted values are generated with exact rational arithmetic by `figures/trace_integrity_fibres.py`.

The labelled inverse from a trace value S is the positive-root equation

$$U^2 - \left(\frac{R}{p}(S - p - a) - 2a \right) U + a^2 = 0, \quad (\text{TR16})$$

followed by the original integer, range and gate tests. Keeping the source word U recovers its factor exponents and tag through TR9. Equal trace values at *different* distinguished denominators a are not merged by TR13.

For arbitrary retained positive source weights w_U , use $\|x\|^2 = \sum_U w_U |x_U|^2$. The exact quotient norm and minimum-norm section are

$$\|z\|_{\text{quot}}^2 = \sum_S \frac{|z_S|^2}{\sum_{S(U)=S} w_U^{-1}}, \quad x_U = \frac{z_{S(U)} w_U^{-1}}{\sum_{S(V)=S(U)} w_V^{-1}}.$$

Cauchy–Schwarz applied on each actual fibre proves the lower bound, and substitution proves equality for the displayed section. In the counting norm $w_U = 1$, it places z_S/m_S on each of the m_S original words, with squared quotient norm $\sum_S |z_S|^2/m_S$. At a two-point fibre this section is not integral when z_S is odd. Choosing one original representative instead gives an integral section with a different norm. No section discards the kernel. These are norms of the specified finite word source, not a replacement of the weighted analytic conductor. Reduction modulo two sends the original middle mass two to zero but retains its two words and the kernel generator. It is not an absent source.

For example, at $p = 1009, a = 253$, the middle pair $U = 11, 5819$ has $S = 2132270$. The exterior word $U = 11099 = p \cdot 11$ has $S = 3906350$ and returns

$$(253, 87032, 3818056).$$

It is integral but is not a canonical M state at the uncorrected word 11099.

5 An integral first trace alone retains a square-part obstruction

Consider arbitrary positive rational tails y, z at the original integer a :

$$\frac{1}{y} + \frac{1}{z} = \frac{R}{pa}, \quad N = y + z \in \mathbb{Z}.$$

This section does not assume the raw full gate of TR7.

Theorem 5.1 (Exact remaining denominator). *The reduced denominators of y, z coincide at d , and*

$$\boxed{d^2 = \frac{R}{\gcd(R, N)}, \quad \text{den}(a^2 + y^2 + z^2) = d^2.} \quad (\text{TR17})$$

In particular squarefree R makes an integral first trace sufficient. Every such pair has the complete integer-factor presentation

$$bc = (pa)^2, \quad b, c > 0, \quad y = (pa + b)/R, \quad z = (pa + c)/R,$$

with

$$\boxed{K(R) \mid pa + b, \quad K(R) = \prod_{\ell \in \parallel R} \ell^{\lceil e/2 \rceil}.} \quad (\text{TR18})$$

The pair is integral precisely when $R \mid pa + b$; in the relaxed pair $d = R/\gcd(R, pa + b)$.

Proof. Since $y + z$ is integral, write $y = Y/d, z = Z/d$ with both numerators coprime to d . The reduced denominator of yz is d^2 . But $yz = Npa/R$ has denominator $R/\gcd(R, N)$, proving the first equality. The second follows from $a^2 + y^2 + z^2 = a^2 + N^2 - 2Npa/R$ and odd R , coprime to pa . Thus $d \mid R$, and $b = Ry - pa, c = Rz - pa$ are positive integers; the reciprocal identity gives $bc = (pa)^2$. They are units modulo R . Conversely every such factor pair gives positive rational tails and

$$y + z = \frac{2pa + b + c}{R} = \frac{(pa + b)^2}{Rb}.$$

Its integrality is $R \mid (pa + b)^2$, exactly TR18. The remaining assertions are the literal denominators of $(pa + b)/R$. $\square$

The failed shell-change attempt is completely concrete at the hard prime 1009:

$$\boxed{a = 286, R = 135, K(R) = 45, b = 11, (x, y, z) = (286, 6413/3, 168238642/3).} \quad (\text{TR19})$$

It satisfies ES, has first denominator trace $56081971 \in \mathbb{Z}$, and second trace $28304240703866897/9$. Its missing denominator is exactly three. Removing its square factor from R gives $R' = 15, a' = 256$. The complete original divisors are $2^f, 0 \leq f \leq 16$, whose residues modulo fifteen are 1, 2, 4, 8. Neither original target, 14 for M or 11 for E, is present. Thus stripping that square does not transport the original factor box to an occupied shell. The smaller control $p = 193, a = 55, R = 27$ similarly gives $(55, 1180/3, 2505140/3)$ with integral first trace, while the compressed shell $R' = 3, a' = 49$ has both gates empty. These are failures of the specified map, not ES counterexamples.

6 The literal coefficient observation and a rationality boundary

For the original literal roots (p, x, y, z) , keep the exact normalization

$$\mathcal{H}(T) = AT^4 + T^3 + BT^2 + CT + D = -S^{-1}(T - p)(T - x)(T - y)(T - z).$$

These are the preceding integral-completion coordinates, not the reciprocal cubic and not the analytic conductor moments [4, IC4–IC5]. The first two traces of multiplication by T on the rank-four root algebra are

$$\boxed{\mathbf{N}_1 = -A^{-1}, \quad \mathbf{N}_2 = A^{-2} - 2B/A.} \quad (\text{TR20})$$

They follow from TR2. For rational ES roots, Corollary 2.2 says that integrality of these two traces is equivalent to the original integer solution, since subtracting p, p^2 gives the denominator traces. On the rank-eight completion obtained by adjoining $\eta^2 = \mathcal{H}'(T)$, the traces of these multiplication operators are *twice* TR20. The two copies and their original basis are retained; a weighted analytic shift-conductor moment is a different observation and has not been substituted for either one.

The two traces also determine the whole unordered denominator polynomial, not merely an integrality predicate. With $s_1 = \mathbf{N}_1 - p$ and $s_2 = \mathbf{N}_2 - p^2$, it is exactly

$$\boxed{g_{p, s_1, s_2}(T) = T^3 - s_1 T^2 + \frac{s_1^2 - s_2}{2} T - \frac{p(s_1^2 - s_2)}{8}.} \quad (\text{TR20a})$$

The reciprocal equation supplies its constant term. Every integral trace pair whose polynomial splits into positive rational roots therefore gives integer roots by Corollary 2.2; in particular $s_1^2 - s_2$ is divisible by eight at odd p . On the original canonical domain, choose the unique root $a \in (p/4, p/2)$. Its remaining p -divisibility pattern selects E (one tail multiple of p) or M (both tails multiples of p), with the M orientation bit retained. The original ordered divisor inverse then recovers the entire box marking. The fibres of the map from original states to $(p, \mathbf{N}_1, \mathbf{N}_2)$ have size one at E and two at M. These claims use only the original distinct-denominator domain; arbitrary input trace pairs still require positivity, rational splitting and the original range test.

Theorem 6.1 (Algebraic targets with every integer trace at every hard prime). *For every prime $p \geq 1009$, $p \equiv 1 \pmod{24}$, there are explicitly constructed positive, distinct algebraic-integer denominators (a, pb, pc) satisfying ES, with $a = (p + 3)/4$, for which b, c are irrational over $\mathbb{Q}$. All power traces are integers. Their literal quartic splits locally over $\mathbb{Q}_p$ and has the same middle root-cluster valuations and negative pair character as an original M state. It also satisfies the inherited scalar height and separation bounds. This family is outside the rational-root domain of Corollary 2.2.*

Proof. There are exactly $(p - 1)/4$ residues $k_0 \pmod{p}$ such that

$$\left(\frac{k_0}{p}\right) = \left(\frac{3k_0 - 1}{p}\right) = -1. \quad (\text{TR21})$$

To count them set $s = 3k_0$, since 3 is a square. The sum $\sum_s \chi(s(s - 1)) = -1$ follows by counting $(2s - 1)^2 - (2y)^2 = 1$: there are $p - 1$ nonzero choices of its first factor, each uniquely determining its second. Expand $(1 - \chi(s))(1 - \chi(s - 1))/4$. The values at zero and one are zero because -1 is a square, so the count is exactly $(p - 1)/4$.

Let k_0 be the least such residue, and set

$$m = (p - 1)/6 = (2a - 2)/3, \quad K_0 = \lfloor (m + 1)^2/2 \rfloor + 1, \quad k = K_0 + ((k_0 - K_0) \bmod p).$$

Let b, c be the two roots of

$$T^2 - 3kT + ak, \quad \Delta = 9k^2 - 4ak. \quad (\text{TR22})$$

The exact inequalities

$$\boxed{(3k - m - 1)^2 < \Delta < (3k - m)^2} \quad (\text{TR23})$$

follow because the first difference is $2k - (m + 1)^2 > 0$ and the second is $4k + m^2 > 0$. The bounding integers are positive. Thus Δ is not a rational square; b, c are irrational algebraic integers. They are positive, $b > a/3$ and $c < 3k$, from $b = 2ak/(3k + \sqrt{\Delta})$. Their reciprocal identity is

$$\frac{1}{a} + \frac{1}{pb} + \frac{1}{pc} = \frac{1}{a} + \frac{3}{pa} = \frac{4}{p}.$$

Their monic cubic is

$$T^3 - (a + 3pk)T^2 + pak(p + 3)T - p^2a^2k. \quad (\text{TR24})$$

All power traces are integers by TR2 and its integral recurrence beyond degree three. This does not make its quadratic factor split over $\mathbb{Q}$. There is an equally literal algebraic version of the raw source:

$$U_{\pm} = \frac{9k - 2a \pm 3\sqrt{\Delta}}{2}, \quad U_+U_- = a^2, \quad a + U_{\pm} = 3b \text{ or } 3c. \quad (\text{TR25})$$

Both $U_{\pm}$ are positive irrational algebraic integers, since they are the roots of $U^2 - (9k - 2a)U + a^2$. They are genuine divisors of a^2 in that quadratic integer ring and pass its sum gate. Also $U_{\pm} < 9k < pa^2$. Substitution in TR7 returns the two tail orientations. This is not the source $\mathbb{Z}_{>0}$: extending its scalar ring has supplied two new words. The field and the conjugation label cannot be discarded.

Modulo p , $a = 3/4$, and $\Delta = 3k(3k - 1)$ is a nonzero square by TR21. Unit-derivative Hensel lifting therefore gives both roots in $\mathbb{Z}_p$. Their product has character $\chi(ak) = -1$. They are distinct units, and neither equals one modulo p : that would force $k = 4/9$, a square. Thus the literal roots p, pb, pc have all pairwise difference valuations one, and the singleton a is separated by units. The derivative valuations are $(2, 0, 2, 2)$. With $S = p + a + 3pk$, the normalized coefficient valuations are $(0, 1, 2, 3)$, by direct expansion. These are the original M cluster conditions from [4, IC7, IC14]; that calculation does not force global rationality. For the lift itself, if $z^2 = \Delta$

(mod p^n), set $z' = z + p^n t$ with $t = -(z^2 - \Delta)/p^n (2z)^{-1} \pmod{p}$; both compositions are checked directly. Standard background is [3, Chapter 7].

Finally

$$k \leq (p+5)^2/72 + p < p^2/60 \quad (p \geq 1009).$$

The last comparison is $p^2 - 410p - 125 > 0$. Put $s_p = (p^2 + 3p)/4$ and $B_p = p + s_p(s_p + 1)$. Then

$$pc < 3pk < p^3/20 < s_p(s_p + 1)/3, \quad S < B_p.$$

In increasing order the literal roots are a, p, pb, pc , with all adjacent gaps larger than one. Their Vandermonde product is consequently larger than twelve, and the product of the three differences from any one root is at least two. The same scalar inequalities as the preceding integer-source bounds follow:

$$\text{Disc}(\mathcal{H}) > 144/B_p^6, \quad 2/S \leq |\mathcal{H}'(\ell_i)| < S^2.$$

No integral-point conclusion is drawn from them. □

At $p = 1009$ the construction gives

$$a = 253, \quad k_0 = 13, \quad k = 15148, \quad \Delta = 2049827360, \quad 45275^2 < \Delta < 45276^2.$$

The quadratic is $T^2 - 45444T + 3832444$. These explicit irrational denominator roots have all integer traces and the required local split pattern. They are not original divisor states. In particular, replacing a rational-root test by an arbitrary finite list of symmetric integer traces is unsound: this example has *every* symmetric power trace integral.

7 What has and has not closed

The global rational-source integrality return is complete. It may be tested by two traces on arbitrary rational ES denominators, or one trace on TR7 with the full integer gate and its distinguished prime bit. The exact coefficient kernel in TR13 also recovers all oriented M states. These maps preserve all original fibre multiplicities.

The remaining assertion is still that every prescribed hard prime produces at least one integral trace in that raw source, equivalently an original opposite divisor pair. No lower bound on $|\Theta_{p,a} \cap \mathbb{Z}|$ or on rank $\ker \Phi$ has been proved that guarantees this for some shell. TR17–19 show why stripping a square from a residual is not an available replacement. Theorem 6.1 shows why the algebraic completion cannot supply the missing rational-root condition by itself. Turn 7 is not marked complete and no witness-guaranteed Turn 8 procedure is asserted.

The executable bounds, all source tables, rejected false implications, and normal-mode, optimized-mode and separate-checker outcomes are recorded separately. Finite scans check the stated formulas and fibres; they do not establish universal occupancy.

References

- [1] Euler, L. (1750/2007). *A double demonstration of a theorem of Newton, which gives a relation between the coefficient of an algebraic equation and the sums of the powers of its roots* (J. Bell, Trans.). Original: *Opuscula varii argumenti*, 2, 108–120, E153. arXiv:0707.0699. §§3, 5–7 are used only for the classical Newton identities, which are also derived in the proof. <https://arxiv.org/abs/0707.0699>
- [2] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94, 50–105. arXiv:1107.1010, §2, Propositions 2.2 and 2.6. The Type-I/II identities are inherited; no averaged theorem is used as a pointwise occupancy statement. <https://arxiv.org/abs/1107.1010>

- [3] Milne, J. S. (2020). *Algebraic number theory*, version 3.08. Chapter 7, Newton’s lemma, supplies the standard local lifting background. The exact quadratic lift used here is given in the proof. <https://www.jmilne.org/math/CourseNotes/ANT.pdf>
- [4] The Clankers. (2026). *Original integral completion of the literal ES –Fabel inverse*. Preserved Turn 7 source, 20 September 2026, `core.tex` in the supplied Turn 7 integral-completion directory. The original IC4–IC7 quartic coordinates, IC14 local residue calculation, and the foreign-prime source-domain obstruction are used at their stated scope. The exact source hashes and retrieval locators are recorded in `source_reading.json`. Earlier work remains separately preserved in the cumulative archive; it is not rewritten as part of this continuation.